



STUDY

European Heat Pump Market 2025–2026

An in-depth market analysis and insight report on the European Heat Pump Market, focusing on the trends, segments, and forecasts for 2025–2026, designed for professional stakeholders.

August 2026

Caspr.

STUDY REPORT

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• Executive Summary

Finding 1 — Installed base far exceeds annual shipments

In 2025, Europe's heat pump installed base is described as significantly larger than annual shipments, indicating a structurally large market despite weaker recent sales. The summary also notes a rebound in total units versus previous years, suggesting recovery in overall market scale.

Finding 2 — Residential leads volume; commercial and industrial offer differentiated value pools

The residential segment dominates due to its large installed base and broad geographic penetration across Europe. Commercial demand is driven by value-rich, large-capacity projects, while the smaller industrial segment is positioned as a decarbonization opportunity with upside potential.

Finding 3 — Market maturity varies sharply by country

Country-level analysis highlights major differences across Europe, with France, Italy, Germany, and Nordic markets such as Norway and Finland showing distinct levels of maturity and policy influence. Residential demand is widespread, whereas commercial and industrial applications are more concentrated in regions with advanced heating systems.

Finding 4 — Policy is the primary swing factor for recovery

The report emphasizes that policy and regulation are central to market dynamics, with a transition from crisis-driven support to more structured electrification strategies. EU-wide coordination and national subsidies are helping recovery, but policy volatility is undermining investor confidence and slowing momentum.

Finding 5 — Competition is concentrated, but industry structure is shifting

Daikin and Mitsubishi are identified as dominant in model registration, pointing to meaningful brand concentration. At the same time, mergers and investments by Asian firms indicate ongoing industrial consolidation and changing competitive dynamics.

Finding 6 — Strong European manufacturing is offset by critical component dependencies

Europe has a robust heat pump manufacturing base, but the supply chain remains vulnerable through dependence on key components such as compressors and electronics. This limits full autonomy and creates strategic risk despite local production strength.

Finding 7 — Technology is aligning with decarbonization and regulatory goals

Innovation is moving toward low-GWP refrigerants, tighter integration with renewable energy systems, and better performance across varying operating conditions. These technology shifts are directly linked to regulatory direction and are shaping future investment and adoption across segments.

Finding 8 — 2030 growth ambitions are at risk without structural fixes

The summary flags a pronounced gap between current stock goals and the annual installation levels required to meet them. Economic pressure, labor shortages, unstable policy, and the need for more renewable capacity all represent material barriers, reinforcing the need for stable long-term frameworks and better industry data.

1

Market Sizing: Installed Base vs. Annual Shipments

1

Market Sizing: Installed Base vs. Annual Shipments

This section distinguishes clearly between installed base, meaning the cumulative stock of operating heat pumps, and annual shipments, meaning units sold or installed within a single year. For Europe in 2025, the stock metric is materially larger than the annual flow metric: the installed base is reported at roughly 28.0 million to 29.3 million units depending on geographic scope and source, while 2025 annual shipments range from 2.34 million to 2.9 million units depending on whether the dataset covers 13 EU member states, 16 mostly residential markets, or 21 European countries.^{10 11 40}

The market picture is therefore not contradictory but scope-dependent. EHPA's broadest 21-country dataset indicates a 2025 rebound after two down years, lifting total stock to 29.3 million units, while narrower Commission and EHPA subsets report lower shipment and stock totals because they exclude some countries and, in some cases, focus on residential market coverage rather than the full heating heat pump universe.^{10 11 13}

1.1

Installed Base Data

Installed base is the stock metric: all heat pumps already operating in the market, not just units sold in the current year. On that basis, Europe's installed base in 2025 is best framed as a range of roughly 28.0 million to 29.3 million units, with about 26 million in the EU alone, because the underlying datasets use different country coverage and product scopes.^{10 40 11}

- EHPA 21-country view: 29.3 million installed units in 2025 across 21 European countries, the broadest headline figure cited for the market.¹⁰
- EHPA 16-country subset: around 28 million installed units after 2025 sales, covering markets that account for over 90% of the market but not all Europe.¹⁴
- EU-only framing: around 28 million in Europe including 26 million in the EU, as used by the Commission's Heat Pump Accelerator Platform.⁴⁰

The growth trajectory is strong even after the 2023 to 2024 slowdown in annual sales. EHPA reported about 20 million installed units after 2022, about 24 million after 2023, and 29.3 million after 2025, implying that the installed base has roughly tripled versus the mid-2010s and more than doubled since 2017 on a pan-European basis.^{32 23 10}

Country concentration remains high within the stock base. France has around 7 million installed heat pumps and Italy just under 5 million, while Norway and Finland lead on household penetration with roughly 650 and just above 540 heat pumps per 1,000 households respectively, showing that volume leadership and penetration leadership are not the same thing.¹⁰

For investors, the key implication is that installed base is the better indicator of service revenue, replacement cycles, and long-run electricity demand impact, whereas annual shipments are the better indicator of current market momentum. The stock base also spans residential, commercial, and some larger building applications more fully than the narrower residential sales datasets often cited in media summaries, so source scope must be checked before comparing figures.^{11 16}

1.2

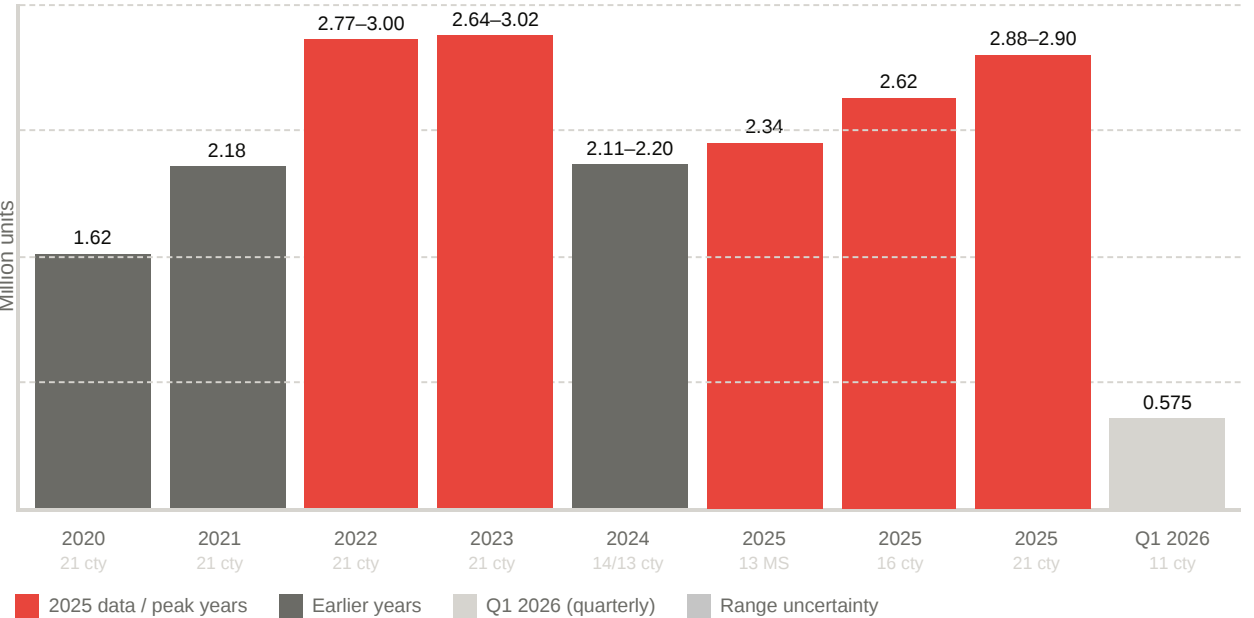
Annual Shipment Data

Annual shipments are the flow metric: units sold or newly installed during a given year. For 2025, the most defensible market framing is a range rather than a single number, because the headline varies from 2.34 million to 2.9 million units depending on country coverage and whether the dataset is a broad Europe total or a narrower subset.^{10 11 14}

Title — European heat pump shipments trend

YEAR	GEOGRAPHY AND SCOPE	SHIPMENTS MILLION UNITS
2020	EHPA 21 European countries	1.62
2021	EHPA 21 European countries	2.18
2022	EHPA 21 European countries	2.77 to 3.00
2023	EHPA 21 European countries	2.64 to 3.02
2024	EHPA 14-country sample or EC 13-member-state sample	2.11 to 2.20
2025	EC 13 member states	2.34
2025	EHPA 16-country subset	2.62
2025	EHPA 21 European countries	2.88 to 2.90
Q1 2026	EHPA 11-country residential sample	0.575

Source: ^{10 32 17 36 11 31}



The historical pattern is now clear. Shipments accelerated sharply in 2021 and 2022, peaked in 2022 on a comparable EHPA basis, fell in 2023, dropped sharply again in 2024, and then rebounded in 2025 by roughly 10% to 13% depending on dataset scope.^{32 36 10 11}

Country-level 2025 shipment leaders were France at 528,000 units and Italy at 423,000 units in EHPA's 21-country dataset. Germany was the standout growth market with 50% year on year growth, Denmark rose 36%, and EHPA states that sales increased in 14 of 18 countries analysed, indicating that the rebound was broad but not universal.^{10 14}

Previously implied expectations for 2025 had been more cautious after the 2024 contraction. Actual 2025 outcomes therefore beat the depressed 2024 run rate, but they still remained below the pace required to align comfortably with Europe's longer-term installation ambitions, especially if the market is to move from roughly 2.9 million annual additions toward 4 million or more by 2030.^{11 40}

1.3

Data Reconciliation and Trends

The main reason figures differ across EHPA, European Commission, and IEA publications is not arithmetic inconsistency but dataset design. EHPA publishes multiple cuts of the market, including a 21-country Europe-wide series and narrower country subsets, while the Commission's 2025 headline of 2.34 million units refers only to preliminary data from 13 EU member states, and IEA generally reuses EHPA and Commission inputs for regional synthesis rather than creating an independent primary sales census.^{10 11 13}

- EHPA 21-country total: Best used for the broadest Europe market sizing, with 2.9 million 2025 shipments and 29.3 million installed stock.¹⁰
- EHPA 16-country subset: Best used for like-for-like trend analysis across the markets covered in the 2025 rebound note, with 2.62 million units sold and around 28 million installed.¹⁴
- European Commission 13-member-state preliminary set: Best used for EU-core policy tracking, with 2.34 million 2025 sales after 2.11 million in 2024.¹¹
- IEA synthesis: Best used for directional interpretation, confirming Europe returned to growth in 2025 and that double-digit growth continued into early 2026.^{13 16}

On trend, the comparable narrative is robust across sources even where absolute totals differ. Europe peaked in 2022, declined in 2023 by about 5% on EHPA's like-for-like series from 2.77 million to 2.64 million, fell again in 2024 by about 21% in EHPA's 14-country sample, then recovered in 2025 by 10% to 13% depending on whether one uses the Commission's 13-country set, EHPA's 16-country subset, or EHPA's 21-country total.^{36 17 10 11}

The 2026 signal is positive but still partial. EHPA reported about 575,000 residential heat pump units sold across 11 European countries in January to March 2026, up 17% from 494,000 in the same country sample a year earlier, while the Commission's Heat Pump Accelerator Platform repeats the same 17% Q1 growth signal; however, this is not yet a full-year all-segment Europe total and should not be annualised mechanically.^{31 40}

For market sizing, the cleanest reconciliation is therefore as follows. Use 29.3 million installed units and 2.9 million 2025 shipments for the broadest Europe-wide headline, use 26 million installed units for the EU-only stock framing, and use 2.34 million to 2.62 million when discussing narrower EU-core or residential-heavy subsets; all are valid if their scope is stated explicitly.^{10 11 40}

2

Segment Breakdown: Residential, Commercial, and Industrial

2

Segment Breakdown: Residential, Commercial, and Industrial

This section assesses the European heat pump market by end-use segment rather than by total market alone. The central analytical constraint is that Europe has relatively robust data for total annual shipments and installed base, but far weaker public disclosure for unit-level shipments split cleanly across residential, commercial, and industrial applications, so segment conclusions must rely more heavily on market value, technology mix, project pipeline evidence, and application economics than on a complete unit census.^{11 58 16}

2.1

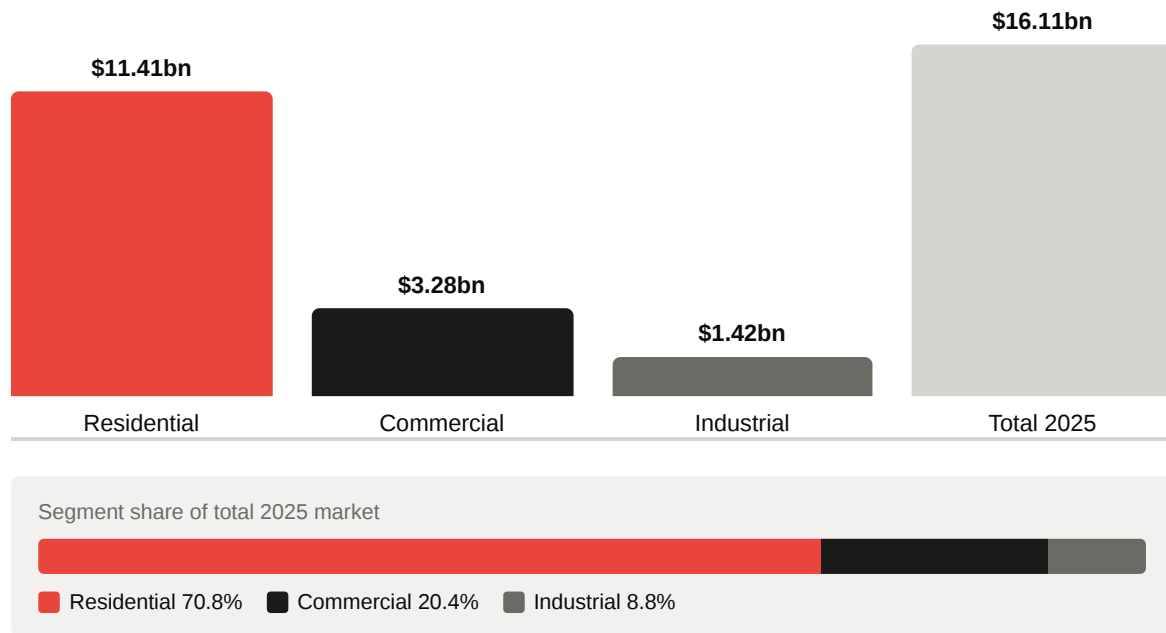
Market Value and Segmentation

Residential remains the economic core of the European heat pump market in 2025 to 2026, reflecting the region’s large installed base in single-family homes, apartments, and small multifamily retrofits. MarketsandMarkets sizes the total Europe heat pump market at USD 16.11 billion in 2025 and indicates a rise to USD 31.12 billion by 2030 at a 14.1% CAGR, with 2026 segment values of about USD 11.41 billion for residential, USD 3.28 billion for commercial, and USD 1.42 billion for industrial applications.⁵⁸

Title — Europe heat pump market by segment

SEGMENT	2026 VALUE USD BN	STRATEGIC POSITION
Residential	11.41	Dominant by revenue and by unit volumes in buildings.
Commercial	3.28	Strong in offices, retail, hospitality, healthcare, and multifamily central systems.
Industrial	1.42	Smallest today, but strategically important for process heat decarbonisation.
Total market 2025	16.11	Baseline market size before forecast expansion to 2030.

Source: ⁵⁸



- **Residential:** This segment dominates because most European heat pump deployments still occur in space heating and domestic hot water for dwellings, especially air-to-water and reversible air-to-air systems. The installed base evidence from EHPA and the Commission also implies that the overwhelming majority of Europe’s cumulative stock sits in buildings rather than industrial process applications, even though those stock datasets are not fully segmented by end use.^{11 73}
- **Commercial:** Commercial demand is more value-dense than unit-dense because projects often involve larger capacities, hydronic systems, and more complex controls. It is strongest in schools, offices, hotels, hospitals, logistics, and mixed-use buildings where electrification, cooling demand, and building energy regulations increasingly converge.^{16 60}
- **Industrial:** Industrial heat pumps remain the smallest segment by current market value, but they have disproportionate strategic importance because they address fossil fuel displacement in low and medium temperature process heat. JRC states that heat pumps capable of reaching up to 100°C could cover up to 11% of Europe’s process heat demand, which supports a structurally higher long-term growth case than current revenue share alone suggests.^{27 16}
- **Geographic dominance by segment:** Residential value is broad-based across France, Italy, Germany, the Nordics, and the UK, while commercial and industrial concentration is higher in countries with stronger district heating systems, larger service-sector building stocks, and more advanced industrial decarbonisation programs. Northern and Central European markets are particularly relevant for larger hydronic, networked, and process-heat applications because policy support and engineering ecosystems are more mature there.^{11 16}
- **Investor implication:** Revenue concentration in residential should not be mistaken for the best marginal growth opportunity. Residential drives scale and replacement cycles, but commercial and especially industrial projects can deliver higher average selling prices, longer service contracts, and stronger policy-linked upside if carbon pricing and industrial support mechanisms deepen through 2030.^{58 27}

2.2

Applications and Technology Integration

Application mix differs materially by segment. Air-source systems dominate residential deployment because they are cheaper and easier to install at scale, while ground-source and water-source systems are more common where higher utilisation, stable source temperatures, or larger centralised loads justify higher upfront capital intensity.^{65 11}

- Air-source heat pumps: These are the volume backbone of the European market and are especially prevalent in residential retrofits and new-build housing. Grand View Research estimates air-source systems accounted for 89.85% of Europe heat pump revenue in 2025, underscoring how strongly the market still tilts toward lower-complexity building applications.⁶⁵
- Ground-source heat pumps: GSHPs remain a smaller share of annual sales but are important in premium residential, campus, and commercial settings where higher seasonal efficiency and load stability matter. Research on shallow geothermal and district heating and cooling indicates that GSHPs become more attractive when integrated with cooling loads and networked thermal balancing, which improves annual extraction potential and system economics.⁵²
- Water-source heat pumps: These are relevant in dense urban projects, waterfront developments, and certain commercial or district energy schemes where rivers, lakes, aquifers, or looped water networks can be used as stable thermal sources. Their market share is smaller than air-source, but they are strategically important in high-load applications where source stability improves performance and capacity utilisation.^{65 11}
- District heating integration: Heat pumps are increasingly being deployed as central assets in district heating and district energy systems rather than only as building-level appliances. IEA notes Europe is advancing large-scale applications in district heating, and recent modelling work finds that long-duration thermal storage attached to district heating networks can reduce long-duration electricity storage needs by an average of 36% in highly electrified European heating systems.^{16 49}
- Commercial integration pathway: In commercial buildings, heat pumps are increasingly paired with building management systems, thermal storage, rooftop solar, and cooling-led electrification strategies. This favors reversible systems and central hydronic architectures that can arbitrage time-of-use tariffs, improve self-consumption, and support demand response.^{16 60}
- Industrial integration pathway: Industrial heat pumps are moving from niche recovery systems toward core decarbonisation equipment for food, paper, chemicals, and other low to medium temperature processes. IEA states that industrial heat pumps below 120°C have entered commercial operation in recent years, while JRC highlights EU technology progress toward higher temperature ranges, expanding the addressable process-heat envelope over time.^{16 27}
- District heating relevance for investors: Network-scale heat pumps create a bridge between the commercial and industrial segments because they combine utility-style project characteristics with building heat demand. This makes them especially important in Nordic and continental European markets where municipal utilities, waste heat recovery, and thermal storage can support larger project sizes and more stable cash flow profiles.^{16 50}

2.3

Segment Analysis and Data Gap Implications

The main data gap for investors is the absence of a harmonised Europe-wide public dataset that reports annual shipments by residential, commercial, and industrial segment on a like-for-like basis across countries. Public sources are much stronger on total annual shipments and installed

base than on segment-level unit flows, which means segment market shares by units cannot be derived with the same confidence as segment market shares by revenue.^{11 73 60}

- Why the gap matters: Without unit-level segmentation, investors cannot cleanly separate volume growth from price and mix effects. A rise in industrial market value, for example, may reflect more megawatt-scale projects and higher system complexity rather than a large increase in unit counts, while residential revenue can soften even if unit shipments recover because average selling prices normalize.^{58 16}
- Installed base versus annual shipments: The distinction is especially important at segment level. Residential has by far the largest installed base and annual shipment count, but commercial and industrial can command higher revenue per installed unit, so unit scarcity in the data tends to understate their economic significance if analysts rely only on shipment headlines.^{11 27}
- Best alternative metrics: Investors should track segment revenue, average project size in kW or MW, announced manufacturing capacity by product class, district heating project pipeline, industrial waste-heat recovery projects, and policy eligibility by application type. These indicators are imperfect, but they are more decision-useful than forcing false precision onto missing unit data.^{16 60}
- Residential proxy set: Use country shipment data, housing stock, retrofit rates, and technology mix such as air-to-water versus reversible air-to-air as the best available proxies for annual residential flow. This is the segment where public shipment data are most visible, but even here cross-country comparability is weakened by differences in product definitions and reporting scope.^{73 11}
- Commercial proxy set: Track non-residential floor area retrofits, public procurement, hotel and healthcare capex, and central plant project announcements. Commercial demand often appears in market value and project databases before it appears in any standardised unit statistics, because systems are bespoke and frequently bundled with controls, storage, and cooling upgrades.¹⁶
- Industrial proxy set: The most informative indicators are process-heat temperature bands, waste-heat availability, carbon cost exposure, and subsidy coverage for electrification. JRC's estimate that up to 11% of Europe's process heat demand could be served by heat pumps up to 100°C provides a better measure of addressable market than unit counts, because industrial systems vary enormously in scale and configuration.²⁷
- Bottom line for market assessment: Segment valuation is credible, but segment unit shipment precision is not yet. For underwriting purposes, residential should be treated as the scale segment, commercial as the systems-integration segment, and industrial as the option-value segment with the highest upside to policy, carbon pricing, and high-temperature technology progress.^{58 16}

3

Country-Level Market Analysis

3

Country-Level Market Analysis

Country performance in Europe’s 2025 heat pump market remained highly uneven, and the distinction between installed base and annual shipments is critical for interpreting leadership. Installed base indicates cumulative market depth and future replacement and service potential, while 2025 shipments indicate current momentum; on that basis, France and Italy remained the largest stock and volume markets, Germany was the clearest rebound market, and Norway and Finland continued to lead on household penetration rather than absolute unit volume.^{10 14 13}

Country analysis also needs to be read through a policy lens. The 2025 recovery was broad, with sales rising in 14 of 18 countries analysed, but the strongest markets were those combining stable subsidies, credible electrification policy, and more favourable relative economics between electricity and fossil heating, while weaker markets were constrained by policy volatility, high electricity-to-gas price ratios, and slower renovation activity.^{81 11 101}

3.1

Market Leaders and Penetration Rates

Volume leadership and penetration leadership diverged sharply in 2025. France remained Europe’s largest market by installed base at around 7 million units and led annual shipments at 528,000 units, while Italy followed with just under 5 million installed units and 423,000 units sold; Germany was smaller in cumulative stock than France or Italy but delivered the strongest large-market acceleration, with 50% year on year growth and heat pumps reaching 50% of national heating generator sales for the first time.^{10 13 14}

Penetration leaders were instead the Nordics. EHPA reports roughly 650 heat pumps per 1,000 households in Norway and just above 540 per 1,000 households in Finland, far ahead of larger continental markets such as France and Italy, confirming that mature electrified heating systems and cold-climate familiarity matter more for penetration than population size alone.^{10 91}

Title — Leading European heat pump markets in 2025

COUNTRY	INSTALLED BASE ESTIMATE	2025 SHIPMENTS	KEY MARKET SIGNAL
France	~7.0 million units	528,000	Largest installed base and annual volume market.
Italy	Just under 5.0 million units	423,000	Large stock base, but growth moderated by weaker construction and subsidy changes.
Germany	Lower than France and Italy; exact 2025 stock not consistently published in the same source set	Not stated in the cited headline source set; growth +50%	Strongest major-market rebound; heat pumps reached 50% of heating generator sales.
Norway	High relative stock	Not a volume leader	Penetration leader at about 650 per 1,000 households.
Finland	High relative stock	Not a volume leader	Penetration leader at just above 540 per 1,000 households.

COUNTRY	INSTALLED BASE ESTIMATE	2025 SHIPMENTS	KEY MARKET SIGNAL
Denmark	Moderate stock	Not stated in the cited headline source set; growth +36%	One of the fastest-growing 2025 markets.

Source: ^{10 14 13}

- France: The market combined scale with maturity, but 2025 momentum was softer than in Germany, and IEA notes that sales declined moderately even as France remained Europe's largest heat pump market. This suggests a market moving from rapid expansion toward a more policy-sensitive replacement and renovation cycle.^{13 10}
- Italy: Italy remained one of Europe's two largest installed-base markets, but 2025 demand was affected by weaker housing construction, slower renovation activity, and the continued adjustment after subsidy changes. For investors, that makes Italy a large but more policy-volatile market than its stock size alone implies.^{10 14}
- Germany: Germany was the standout turnaround case in 2025. The market recovery was driven by strengthened subsidies and improved confidence after prior legislative uncertainty, and the fact that heat pumps overtook gas boilers is strategically important because it signals a structural shift in the new heating market rather than a one-off cyclical rebound.^{14 13 10}
- UK: The UK remained a lower-penetration market than continental leaders, but policy-backed growth has improved its trajectory. Official UK investor materials describe the country as one of Europe's fastest-growing heat pump markets, though from a comparatively small installed base and with persistent affordability barriers linked to power pricing.^{100 102}
- Norway and Finland: These are the benchmark penetration markets, reflecting long-standing electrified heating traditions, cold-climate suitability, and supportive economics. Their strategic relevance is less about incremental unit volume and more about technology validation, premium product mix, and lessons for high-renovation, high-electrification pathways elsewhere in Europe.¹⁰

11

3.2

Emerging Markets and Potential

The highest structural upside now sits in lower-penetration Central and Eastern European markets rather than in the already-saturated Nordic leaders. Poland, Czechia, Slovakia, Hungary, Romania and parts of the Baltics generally start from a much lower installed base per household, but they combine large fossil-heated building stocks, significant renovation needs, and rising policy attention, which creates a strong medium-term catch-up opportunity if economics improve.^{11 101 97}

Czechia stands out as an early growth signal within the region, with EHPA reporting 31% sales growth in 2025, while Poland is repeatedly highlighted in Commission and BUILD UP material as a market with stabilisation potential after subsidy-programme disruption. These markets are not yet penetration leaders, but they are among the most important for Europe's next phase of volume expansion because their heating systems remain more carbon-intensive and less electrified than those in the Nordics or France.^{15 40 101}

- Growth potential: Central and Eastern Europe offers the largest addressable retrofit pool because many homes still rely on coal, gas, or oil systems and have comparatively low heat pump penetration. That creates a larger theoretical runway for annual shipments growth than in already mature markets where installed base is high and incremental adoption is harder.^{97 11}

- Key barriers: The main bottlenecks are high upfront cost relative to household income, weak or unstable subsidy administration, installer shortages, and unfavourable electricity pricing relative to gas or solid fuels. In Poland specifically, the Commission's Heat Pump Accelerator Platform notes that market stabilisation requires predictable subsidies, certified installers, a national strategy, and a stronger information campaign after disinformation and programme challenges.^{40 14}
- Building stock constraint: Older and less efficient buildings raise system-sizing and retrofit costs, which can weaken payback even where subsidies exist. This matters especially in Central and Eastern Europe because a larger share of the housing stock requires envelope upgrades or emitter replacement before heat pumps can perform optimally.^{99 11}
- Commercial and industrial angle: Emerging markets should not be viewed only through residential adoption. District heating modernisation, municipal buildings, and industrial waste-heat recovery can create larger-ticket commercial and industrial heat pump opportunities in countries where household adoption remains slow, particularly where EU funds and industrial decarbonisation programmes can de-risk early projects.^{16 11}
- Investor implication: The region offers higher growth optionality than Western Europe, but execution risk is also higher. Market entry is therefore more attractive through channels tied to subsidy administration, installer networks, public-sector projects, and hybrid renovation packages rather than through pure consumer hardware exposure alone.^{101 40}

3.3

Policy Drivers and Market Influences

The 2025 country pattern confirms that policy quality mattered more than climate or income alone. Markets with stable subsidies, clear decarbonisation signals, and credible restrictions on fossil heating generally outperformed, while markets exposed to abrupt subsidy changes, legislative delays, or poor electricity-to-gas economics underperformed even when their long-term fundamentals remained strong.^{14 11 16}

Germany's rebound was closely tied to stronger support under the federal buildings-efficiency framework and improved confidence after prior policy turbulence. France remained structurally important but faced weaker momentum during subsidy redesign, Italy was affected by the wind-down of earlier generous tax incentives and softer construction activity, and the UK's growth case continued to depend heavily on grant support because retail electricity remains expensive relative to gas.^{14 83 100 102}

- Germany: The key 2025 driver was strengthened subsidy support and clearer market signalling after the earlier heating-law controversy. Househeating Pulse reports support levels reaching up to 65% of cost in Germany, while EHPA and IEA both link the 2025 rebound directly to stronger incentives and restored confidence.^{83 10 13}
- France: France benefits from a large installed base, mature installer ecosystem, and long-running support architecture, but 2025 showed that even large markets can stall when subsidy rules are revised mid-cycle. The policy lesson is that continuity matters as much as generosity for sustaining annual shipments.^{13 85}
- Italy: Italy's market remains highly sensitive to fiscal policy because earlier growth was amplified by generous building-renovation incentives. As those incentives were reduced or normalised, annual shipments became more exposed to underlying construction conditions and household economics, creating a more volatile demand profile despite the country's very large installed base.^{14 10}

- UK: The UK policy model is grant-led rather than penetration-led. GOV.UK states that from July 2026 grants for qualifying air-to-water and ground-source systems replacing oil or LPG rise to £9,000, but the Climate Change Committee still identifies the UK as having one of Europe's highest electricity-to-gas price ratios and one of the lowest heat pump market shares, showing that subsidy support alone does not fully offset distorted running-cost signals.^{100 102}
- Nordic markets: Norway and Finland benefit from structurally stronger electrification economics, high consumer familiarity, and policy environments that have supported heat pumps over many years rather than through short-lived stimulus. Their experience suggests that energy tax structure and electricity affordability are as important as direct capital subsidies in sustaining high penetration.^{10 11}
- Cross-country conclusion: The strongest 2025 markets were those where three conditions aligned: stable subsidies, credible fossil-heating phase-down signals, and electricity pricing that did not excessively penalise electrification. Where one of those three was missing, annual shipments weakened even if long-term installed-base potential remained intact.^{14 11 16}

4

Policy and Regulatory Landscape

4

Policy and Regulatory Landscape

Europe's heat pump policy environment in 2025 to 2026 is being shaped by a more explicit shift from short term crisis support toward structural electrification policy, but market outcomes still depend heavily on whether regulation is stable enough to support multi year investment decisions. For investors, the key distinction is that policy affects annual shipments immediately through grants, taxes, and permitting, while installed base growth depends on whether those measures remain credible long enough to sustain replacement cycles, installer training, and manufacturing scale up.^{112 119 113}

At EU level, the policy stack now combines deployment coordination, industrial support, and energy system reform, while national governments continue to determine the actual economics seen by households, commercial building owners, and industrial users. The result is a two speed landscape: EU institutions are improving strategic direction, but country level subsidy design, electricity taxation, and legislative consistency still determine whether annual shipments recover durably across residential, commercial, and industrial segments.^{112 125 16}

4.1

EU-Level Regulatory Frameworks

EU policy became more interventionist in 2025 to 2026 as the Commission moved from general decarbonisation targets toward implementation platforms and resilience measures directly relevant to heat pumps. The Heat Pump Accelerator Platform was launched in January 2025, its first meeting gathered more than 80 participants, and the Commission now states the platform has 91 members tasked with tracking deployment data, assessing national frameworks, and shaping a roadmap for financing, training, and research needs.¹¹²

AccelerateEU, presented on 22 April 2026, broadened the policy rationale from climate and energy security to affordability and resilience. The Commission frames it as a toolbox to reduce exposure to volatile fossil fuel markets, and the associated Energy Union communication explicitly links faster electrification, including heat pumps, to lower fossil fuel dependence and stronger system resilience.^{119 116 125}

The Heat Pump Accelerator Platform also gives the clearest current EU deployment benchmark. The Commission states that around 28 million heat pumps are installed in Europe, including 26 million in the EU, and that annual installations need to rise from about 2.8 million in 2025 to 4 million by 2030, which is a flow target rather than an installed base target.¹¹²

Industrial policy support is also widening beyond residential buildings. The Commission's Innovation Fund launched the IF25 Heat Auction in December 2025 as the EU's first auction dedicated to industrial process heat decarbonisation, and EHPA's April 2026 manifesto argues that this pilot should become a structural instrument under the Decarbonisation Bank to support industrial heat pumps on a long term basis.^{120 113}

The remaining EU level weakness is that many of the most important price signals still sit outside heat pump specific policy. arXiv based system studies and EHPA's 2026 manifesto point in the same direction: carbon pricing, electricity taxation, and tariff design materially affect heat pump competitiveness, especially where electricity remains taxed more heavily than gas.^{75 74 113}

4.2

National Policy Instruments

National policy remains the decisive driver of annual shipments because grants, tax credits, and retail energy price structures determine project payback at the point of sale. Across Germany, France, Italy, the UK, and Nordic markets, the strongest demand outcomes have generally come where direct support is paired with favourable electricity economics and a credible phase down of fossil heating rather than with subsidies alone.^{11 112 16}

Title — National heat pump policy instruments in key markets

MARKET	MAIN INSTRUMENT IN 2026	POLICY SIGNAL	MARKET IMPLICATION
Germany	KfW heating support under revised BEG rules.	Support remained high in 2025, then KfW announced new conditions from 21 July 2026, with examples showing up to 80% support in some cases before the basic subsidy falls to 15%.	Strong near term demand support, but rule changes reinforce sensitivity to design complexity and timing.
France	MaPrimeRénov' and related renovation support.	France Rénov' confirms MaPrimeRénov' remains the core national renovation subsidy, while France is also introducing from 1 September 2026 an approval mechanism steering support toward qualifying heat pumps made in Europe.	Large market with durable support architecture, but increasing policy conditionality and administrative complexity.
Italy	Reduced renovation tax incentives after Superbonus wind down.	The Commission's 2025 country assessment states Superbonus was reduced to 70% from January 2025, marking a clear retreat from earlier exceptionally generous support.	Demand became more exposed to underlying construction and household economics, increasing volatility in annual shipments.
United Kingdom	Boiler Upgrade Scheme grants.	Ofgem states grants of £9,000 for eligible off gas grid properties replacing oil or LPG with air to water or ground source heat pumps apply from 21 July 2026 to 31 March 2027, and legislation extends the scheme to 2030.	Clear grant support improves affordability, but high electricity to gas price ratios still constrain mass market economics.
Nordic markets	Lower carbon intensity power systems, selective grants, and more favourable tax structures.	Norway's Enova continues targeted support for some heat pump related measures, while Finland's tax tables show low electricity tax for category II users and explicit fuel taxation for gas and other fuels.	Stronger structural economics reduce dependence on stop start subsidy cycles and support higher penetration.

Source: 137 160 164 177 133 145 162 159

- Germany: Germany's framework remains the most consequential for investor sentiment because it combines large market size with strong policy leverage. KfW's July 2026 adjustment shows that support can still be generous, but frequent redesigns create execution risk for installers, distributors, and manufacturers even when the headline subsidy remains attractive.^{137 140}
- France: France retains one of Europe's most institutionalised support systems through MaPrimeRénov', which supports renovation measures including heat pumps. The new ADEME

linked approval regime effective 1 September 2026 adds an industrial policy layer by steering support toward approved European made models, which may strengthen local value chains but also adds compliance and product eligibility complexity.^{160 164}

- Italy: Italy illustrates the downside of overreliance on exceptionally generous fiscal incentives. The reduction of Superbonus to 70% from January 2025, as cited in the Commission’s country documentation, supports the broader market evidence that annual shipments became more volatile once extraordinary support was withdrawn.¹⁷⁷
- United Kingdom: The UK remains a grant led market rather than a structurally low cost electrification market. The Boiler Upgrade Scheme provides a clear capital subsidy path and was extended to 2030, but the need for temporary uplifts for off gas grid homes underlines that running cost economics remain weaker than in leading continental or Nordic markets.^{133 145}
- Nordic markets: Norway and Finland show that policy effectiveness is not only about grants. Norway’s Enova support is targeted rather than blanket, while Finland’s tax structure shows relatively low electricity taxation for certain users and explicit taxation of fossil fuels, supporting the broader conclusion that energy tax design can sustain heat pump competitiveness more durably than short lived purchase subsidies.^{162 159 74}
- Cross segment relevance: These instruments affect residential annual shipments most visibly, but they also shape commercial and industrial adoption through electricity pricing, building renovation support, and industrial decarbonisation eligibility. In particular, industrial heat pump uptake depends less on household style grants and more on whether national and EU schemes treat process heat electrification as a bankable long duration policy priority.^{120 113 16}

4.3

Policy Stability and Recommendations

The main policy lesson from 2023 to 2026 is that instability damages annual shipments faster than it damages long term decarbonisation rhetoric. EHPA’s April 2026 manifesto explicitly calls for long term policy certainty, streamlined legislation, corrected price signals, and structural support for industrial heat pumps, reflecting the sector’s view that abrupt redesigns in subsidies and product rules were a major cause of the 2023 to 2024 market contraction.¹¹³

Italy’s Superbonus wind down and Germany’s repeated redesign of heating support are the clearest recent examples of policy volatility affecting market confidence. Even where support remains substantial, sudden changes in eligibility, timing, or administrative process can delay consumer decisions, disrupt installer pipelines, and raise inventory risk for manufacturers and distributors.^{177 137 113}

For long term market certainty, the most important recommendation is to shift from discretionary stop start subsidy cycles toward rules based frameworks with published multi year trajectories. That means fixed degression schedules for grants, stable boiler phase down timelines, and advance notice periods long enough for manufacturers, installers, and financiers to plan capacity across residential, commercial, and industrial segments.^{113 112}

A second priority is correcting energy price signals. EHPA argues that heat pumps are most competitive when electricity prices are no more than about twice gas prices, while in many countries electricity is taxed three to seven times more heavily than gas; arXiv based research on carbon pricing and tariff design supports the broader conclusion that tax and tariff reform is essential if electrification is to scale efficiently and fairly.^{113 75 74}

A third recommendation is to treat industrial and large commercial heat pumps as a distinct policy class rather than an extension of residential support. The IF25 Heat Auction is a useful start, but the market needs recurring auction rounds, clearer eligibility for process heat and district heating applications, and financing structures that recognise longer development cycles and larger project sizes.^{120 113}

A fourth recommendation is better data governance. The Heat Pump Accelerator Platform is already tasked with tracking deployment data and assessing national frameworks, and this should be expanded into harmonised reporting on annual shipments, installed base, and segment splits so that policymakers and investors can distinguish clearly between stock growth and yearly flow across residential, commercial, and industrial markets.¹¹²

The investor conclusion is straightforward. Europe now has a stronger strategic policy architecture than it did in 2024, but long term market certainty will depend less on new announcements than on whether governments can keep support predictable, simplify compliance, and align electricity taxation with electrification goals over several years rather than several quarters.^{119 113 16}

5

Competitive Landscape and Key Manufacturers

5

Competitive Landscape and Key Manufacturers

Competition in the European heat pump market is increasingly shaped by three overlapping forces: brand concentration at the model-registration level, widening performance dispersion on seasonal efficiency, and a new phase of industrial consolidation around European manufacturing footprints. Publicly visible EPREL model counts suggest a concentrated top tier led by Daikin, Mitsubishi Electric, and Johnson Controls Hitachi, while JRC and Commission sources indicate that a relatively small group of manufacturers captures most sector revenue and that European production remains structurally important despite rising Asian participation through local assembly and acquisitions.^{190 191 38 198}

For investors, the key analytical caution is that EPREL model share is not the same as annual shipments or installed base. It is best read as a proxy for product breadth, regulatory presence, and channel intensity rather than unit sales, whereas M&A and factory investment are better indicators of future competitive positioning across residential, commercial, and industrial segments.^{200 191 24}

5.1

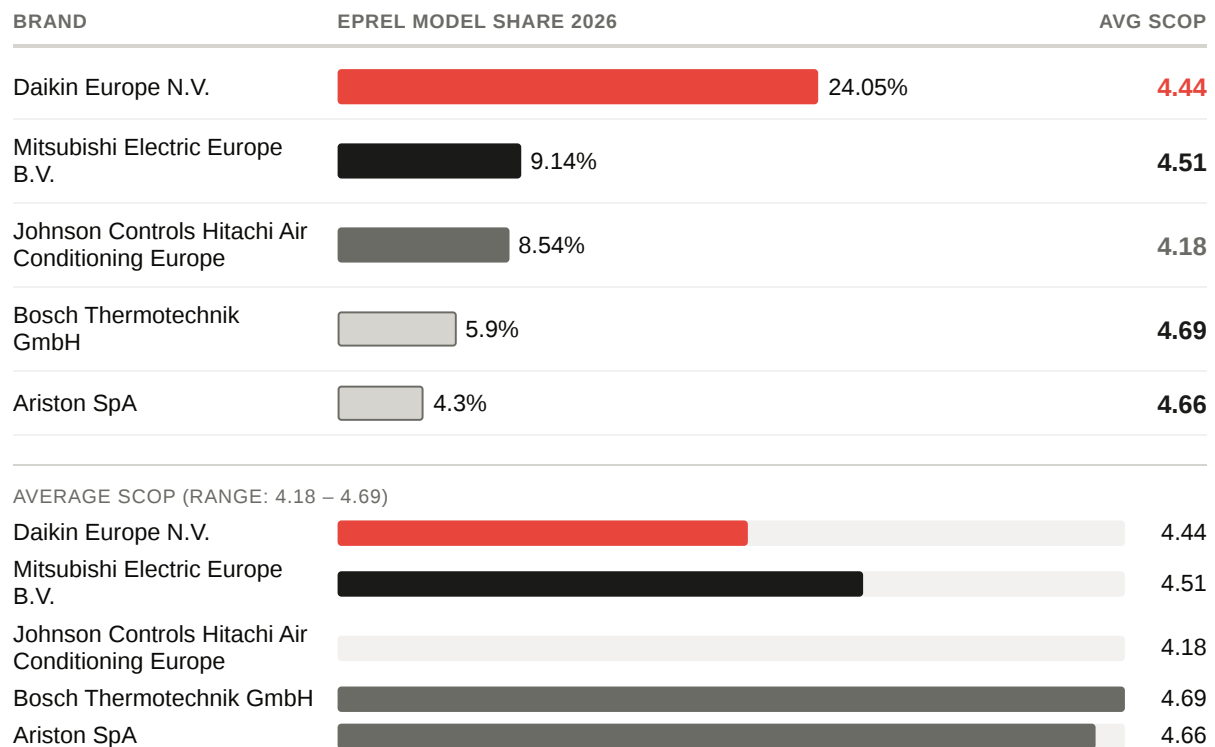
Market Share and Efficiency Rankings

EPREL-based brand-share analysis for Europe in mid 2026 shows a clear top tier by registered model count. Daikin Europe N.V. leads with 14,668 listed models and 24.05% share, followed by Mitsubishi Electric Europe B.V. at 9.14% and Johnson Controls Hitachi Air Conditioning Europe at 8.54%, giving the top three a combined 41.73% share of listed models.^{190 191}

Title — Europe heat pump brand share and SCOP snapshot

BRAND	EPREL MODEL SHARE 2026	AVERAGE SCOP
Daikin Europe N.V.	24.05%	4.44
Mitsubishi Electric Europe B.V.	9.14%	4.51
Johnson Controls Hitachi Air Conditioning Europe	8.54%	4.18
Bosch Thermotechnik GmbH	5.9%	4.69
Ariston SpA	4.3%	4.66

Source: ^{191 190}



The efficiency ranking is more fragmented than the model-share ranking. Bosch at 4.69 average SCOP and Ariston at 4.66 outperform the three largest brands on this metric, while Mitsubishi Electric at 4.51 also exceeds Daikin's 4.44 and Johnson Controls Hitachi's 4.18, indicating that product breadth and seasonal efficiency leadership do not fully coincide.^{190 191}

- Interpretation: EPREL share reflects registered model breadth, not annual shipments or installed base, so it should not be used as a direct sales-share proxy.^{200 191}
- Competitive signal: Large incumbents dominate shelf presence and channel coverage, but smaller-share brands can differentiate through higher average SCOP and likely stronger positioning in premium retrofit or low-temperature hydronic niches.¹⁹⁰
- Revenue concentration: JRC reports that roughly 15 companies account for about 80% of industry revenue, which is consistent with a market that appears broad at the model level but remains economically concentrated at the manufacturer level.²⁴
- Segment caveat: These rankings are most informative for residential and light commercial packaged products; they are less representative of industrial heat pumps, where project engineering, temperature capability, and integrator relationships matter more than model count.³⁸

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5.2

Key Manufacturers and Market Strategies

The competitive field is split between European-rooted manufacturers with dense local production and installer networks, and Asian groups that increasingly compete through European subsidiaries, acquisitions, and localized manufacturing. Commission and JRC sources indicate that Europe retains a strong domestic industrial base, with more than 170 manufacturing sites in earlier Commission mapping and domestic content estimates ranging from 55% to 60% of value in CE Delft work to as high as 73% in JRC analysis.^{193 38 198}

- Daikin: The company's leading EPREL model share points to the broadest product portfolio and strongest regulatory shelf presence in Europe. Its advantage is scale across residential and commercial HVAC, deep distribution, and the ability to spread refrigerant transition and compliance costs across a very large installed channel base.^{191 190}
- Mitsubishi Electric: Mitsubishi combines a top-tier model share with stronger average SCOP than the market leader, supporting a strategy centered on performance credibility and cold-climate reputation. That positioning is particularly relevant in premium residential and light commercial applications where seasonal efficiency and operating cost claims influence conversion rates.¹⁹⁰
- Johnson Controls Hitachi: The group remains a major top-three player by model count, but its lower average SCOP in the cited dataset suggests a broader portfolio mix rather than pure efficiency leadership. Its competitive strength is likely to remain strongest where commercial HVAC relationships and building systems integration matter as much as appliance-level efficiency.^{191 38}
- Bosch and Ariston: Both brands rank below the top three on model share but above them on average SCOP in the cited 2026 snapshot. This supports a strategy of competing on hydronic performance, premium efficiency, and European brand trust rather than on maximum portfolio breadth alone.¹⁹⁰
- NIBE, Stiebel Eltron, and other European specialists: These firms are strategically important beyond visible model-share tables because they are strong in ground-source, hydronic, and higher-specification systems, including parts of the commercial and industrial value chain. Commission material notes 17 EU manufacturers of industrial heat pumps, plus 8 in Norway, underscoring Europe's continued strength in higher-end and industrial applications.¹⁹⁸
- Asian challengers with European localization: Midea and Hisense illustrate the current playbook of entering Europe not only through exports but through acquired brands, local factories, and regional R&D. This reduces tariff and logistics risk, improves subsidy eligibility in some markets, and narrows the perceived gap between imported and European-made systems.^{210 212 38}

The strategic implication is that competition is no longer a simple Europe versus Asia binary. The more relevant divide is between firms with localized manufacturing, installer access, and regulatory agility, and those without those capabilities, regardless of ultimate corporate headquarters.^{24 198}

5.3

Mergers and Acquisitions

Recent consolidation has been less about removing large direct competitors and more about acquiring European brands, production assets, and route-to-market capabilities. The most important pattern is that Asian groups have used acquisitions to localize their European footprint, while European specialists have used deals to deepen technology coverage and strengthen positions in specific geographies such as Germany and the Nordics.^{209 211 206}

- Midea and Clivet: Midea agreed in 2016 to acquire 80% of Clivet, then used the platform to build a localized European HVAC and heat pump base. JRC notes that Midea subsequently relocated domestic hot water heat pump manufacturing from Asia to Italy in 2017, and Midea's later completion of the ARBONIA climate acquisition further broadened its European manufacturing and distribution footprint.^{209 38 213}
- Stiebel Eltron and Danfoss Varmepumpar Thermia: Stiebel Eltron finalized the acquisition of Thermia from Danfoss in 2018, strengthening its position in Sweden, Finland, and Norway and adding ground-source expertise. The often-cited EUR 450 million investment and plan to

quadruple output by 2027 refer to Stiebel Eltron's later organic capacity expansion rather than to the acquisition price itself, so these should be treated as related but distinct competitive events.^{221 223}

- Hisense and Gorenje: The European Commission cleared Hisense's acquisition of Gorenje in 2018, giving Hisense a recognized European appliance brand and manufacturing base. Hisense Europe now reports three major production centers and five R&D centers in Europe, showing how the deal evolved from ownership change into a broader regional industrial platform.^{211 207 212}
- NIBE and Waterkotte: NIBE acquired German manufacturer Waterkotte in March 2020, explicitly to strengthen its position in Germany and add expertise in larger-capacity projects and integrated sustainable energy systems. This is strategically relevant because Germany remains one of Europe's most important growth and policy-driven heat pump markets.^{206 215}

These deals have three market effects. First, they raise barriers to entry by combining brands with local factories and installer channels; second, they accelerate technology transfer into Europe-based product lines; third, they make competitive positioning more regional and segment-specific, especially in hydronic residential systems, commercial HVAC, and industrial or higher-capacity applications.^{38 24}

Data gap note: reliable public sources for recent Europe-wide manufacturer annual shipment share by company remain limited, and the strongest current public comparison is EPREL model share rather than audited unit sales. For that reason, competitive conclusions here should be read as a synthesis of model presence, efficiency positioning, manufacturing footprint, and M&A activity rather than a precise annual shipments league table.^{200 191}

6

Supply Chain and Manufacturing Dynamics

6

Supply Chain and Manufacturing Dynamics

Europe’s heat pump supply chain remains stronger than many other clean technology value chains, but it is not fully vertically integrated. The current manufacturing base is broad in final assembly and selected components, while strategic dependence persists in power electronics, semiconductors, and parts of the compressor chain, creating a mismatch between Europe’s strong market position and its weaker control over critical subcomponents.^{24 204 16}

For investors, the key distinction is between assembly capacity and component sovereignty. Europe has meaningful domestic manufacturing depth across residential, commercial, and parts of the industrial heat pump chain, but the most disruption-sensitive nodes remain imported or globally concentrated, especially compressors, electronic components, and low-technology chips that sit outside the main focus of the European Chips Act.^{24 204}

6.1

Manufacturing Capacities and Employment

The EU manufacturing footprint is larger than the earlier headline of about 170 factories now commonly cited in older policy material. JRC’s 2025 status report identifies at least 266 EU factories involved either in heat pump assembly or in relevant components such as heat exchangers, compressors, and valves, spread across 21 Member States, indicating that the industrial base has broadened materially since the earlier 170-factory mapping.^{24 260}

Title — EU heat pump manufacturing footprint

INDICATOR	LATEST FIGURE	INTERPRETATION
EU factories in assembly or key components	At least 266	Broader industrial base than older 170-factory estimate.
Member States hosting at least one factory	21	Manufacturing is geographically diversified, not concentrated in one market.
EU heat pump sector employment in 2023	About 416,000	Employment has expanded materially from the 2020 baseline.
EU heat pump sector employment in 2020	Nearly 320,000	Useful baseline for measuring post-crisis industrial scaling.

Source: ^{24 260}



- **Employment:** The earlier baseline of nearly 320,000 EU jobs in 2020 remains valid for historical comparison, but JRC's 2025 report shows employment in the heat pump sector rising to roughly 416,000 by 2023, implying that industrial scaling continued even through the 2023 to 2024 market downturn.^{260 24}
- **Geographic clusters:** JRC identifies notable assembly and component clusters in northern Italy, southern and western Germany, and parts of Central Europe. This supports the view that Europe's manufacturing base is not only large but regionally networked, which matters for logistics, supplier co-location, and installer support.²⁴
- **Country contribution:** Italy generated the highest direct gross value added in the EU heat pump industry at EUR 7.6 billion in 2023, followed by France at EUR 4.5 billion and Germany at EUR 2.8 billion. JRC also notes that Germany, Spain, and Portugal were among the countries where created value increased significantly between 2022 and 2023, reinforcing the importance of Iberia as a growing manufacturing platform rather than only an end market.²⁴
- **Segment relevance:** Most public factory counts are dominated by residential and light commercial equipment, but the same industrial ecosystem also supports larger hydronic, district heating, and industrial heat pump applications through shared component and engineering capabilities. That means factory counts should not be read as a residential-only indicator, even if unit output is skewed toward smaller systems.²⁴
- **Capacity caveat:** Public sources remain much stronger on site counts and employment than on nameplate unit capacity by plant. As a result, investors can infer breadth and industrial commitment with confidence, but not a precise EU-wide annual production ceiling by segment from public data alone.^{24 253}

6.2

Domestic Content vs. Import Dependencies

Europe's heat pump value chain is meaningfully local, but not self-sufficient. The most credible public range remains about 55% to 60% domestic content from CE Delft-type assessments versus as high as 73% in JRC analysis, with the difference largely reflecting methodology, scope,

and whether upstream materials, installation, and services are counted alongside manufacturing value added.^{204 250 24}

- Domestic content interpretation: The lower 55% to 60% range is the more conservative investor assumption for hardware-centric analysis, while the higher JRC estimate better captures the broader European value chain including local assembly, installation, and associated services. The practical conclusion is that Europe retains majority local value capture, but not enough to eliminate strategic import risk.^{204 24}
- Compressor exposure: JRC states that European compressor manufacturers include Danfoss, Bitzer, Tecumseh, Copeland, and Dorin, so the EU is not absent from this node. However, it also notes that large Asian firms such as Hitachi and Daikin produce compressors in Japan and Mitsubishi Electric in Thailand, confirming that a meaningful share of the compressor chain remains externally exposed even where raw materials like steel, aluminium, copper, plastics, and rubbers are largely available within Europe.²⁴
- Inverter and electronics exposure: CE Delft identifies compressors, electronic components, and refrigerants as key elements whose production is dominated by a relatively small number of suppliers in a few countries. The same study also flags low-technology semiconductor chips as an under-addressed vulnerability, noting that the European Chips Act focuses more on high-end chips than on the lower-end electronics essential for heat pump controls and power electronics.²⁰⁴
- Trade signal: The European Commission reported that the EU heat pump trade balance swung to a deficit of EUR 390 million in 2021, versus a surplus of EUR 202 million five years earlier, while imports from China doubled in 2021 to EUR 530 million. That does not imply Europe lacks manufacturing, but it does show import intensity rising faster than domestic component substitution in parts of the chain.²⁵⁰
- Strategic vulnerability: The highest-risk chokepoints are not sheet metal, piping, or final assembly, but the electronically intensive and precision-engineered subsystems that can halt production if unavailable. Compressors, inverters, control boards, and low-end chips therefore matter more to resilience than the headline share of local assembly would suggest.^{204 24}
- Segment implications: Residential products are more exposed to standardized imported electronics and compressor modules, while commercial and industrial systems add engineering customization that raises local value capture. Even so, all three segments remain vulnerable to the same upstream electronics bottlenecks because controls and variable-speed drives are increasingly central to performance and grid flexibility.^{204 16}

6.3

Growth Predictions and Strategic Outlook

The medium-term manufacturing outlook is positive, but it is conditional on policy credibility and component localization. IEA's 2026 outlook indicates that under stated policies, heat pump manufacturing capacity rises by about 40% in the European Union by 2035, which is meaningful growth but still slower than the expected expansion in the United States and below China's scale advantage.^{255 16}

- NZIA effect: The Net Zero Industry Act improves the investment case mainly through faster permitting, strategic project treatment, and a clearer industrial policy framework rather than through direct demand creation alone. Its importance is therefore highest for upstream component plants and large-scale manufacturing expansions that need long lead times and regulatory certainty.^{255 258}

- Made in Europe linkage: IEA notes ongoing discussions in France on linking heat pump subsidies to products made in Europe, which signals a broader shift from pure deployment policy toward deployment-plus-industrial-policy design. If replicated more widely, this could improve utilization rates for EU factories, but it may also raise compliance complexity and intensify trade scrutiny.²⁵⁵
- Country outlook: France, Italy, Portugal, and Spain are strategically important not only because of current production presence but because they offer different industrial roles. France and Italy combine scale, brands, and installed supplier ecosystems, while Portugal and Spain appear increasingly relevant as lower-cost manufacturing and value-added growth locations within the EU footprint.²⁴
- What must localize next: The highest-value strategic move is not simply more final assembly lines. It is deeper European capability in compressors, inverters, control electronics, and low-technology chips, because these are the nodes where import disruption can still constrain output even when local assembly capacity exists.^{204 24}
- Investor view: The most attractive manufacturing exposures are likely to be firms and assets that combine EU assembly with partial vertical integration in critical components, or that supply enabling subsystems to residential, commercial, and industrial heat pump platforms simultaneously. Pure assemblers remain exposed to margin pressure if Asian component suppliers retain cost leadership and Europe does not close the electronics gap.^{253 255}
- Bottom line: Europe has enough industrial depth to remain a major heat pump manufacturing region through 2030, but not enough component autonomy to treat the supply chain as secure. The strategic race is therefore shifting from market expansion alone to selective re-shoring of the most disruption-sensitive subcomponents under NZIA-style industrial policy.^{24 204 255}

7

Technology Trends and Innovation

7

Technology Trends and Innovation

Technology innovation in the European heat pump market is now being driven less by basic unit proliferation and more by three technical shifts: refrigerant transition, deeper system integration, and performance optimization under real operating constraints. Across residential, commercial, district heating, and industrial applications, the innovation frontier is moving toward low-GWP refrigerants, grid-interactive and renewable-coupled architectures, and higher seasonal efficiency with lower acoustic and cold-weather penalties, while larger industrial systems are extending usable supply temperatures into the 140 to 200°C development range.^{276 284 278}

For investors, the key implication is that technology leadership is becoming more application-specific. Residential competition is increasingly shaped by refrigerant compliance, SCOP, and noise, while commercial, district, and industrial value creation depends more on controls, thermal storage, source integration, and the ability to deliver higher temperatures and flexible operation within constrained electricity systems.^{276 50 263}

7.1

Refrigerants and Regulatory Drivers

The refrigerant transition is now a central product-development driver in Europe, with the EU F-Gas Regulation accelerating the move away from higher-GWP fluorinated gases and tightening both product placement and servicing rules over time. Regulation (EU) 2024/573 prohibits, from 12 March 2025, the export of stationary air-conditioning equipment and stationary heat pumps covered by Annex IV that contain or rely on fluorinated greenhouse gases with GWP of 1,000 or more, and from 1 January 2026 prohibits the use of fluorinated greenhouse gases with GWP of 2,500 or more for maintenance or servicing of heat pumps and air-conditioning equipment.^{284 283}

- Technology direction: The JRC's 2025 EU heat pump status report identifies refrigerants as a core technology-development theme and notes that the market is moving toward natural refrigerants under regulatory pressure and product redesign requirements.²⁷⁶
- R290 positioning: Propane-based R290 systems are gaining strategic importance because they combine ultra-low GWP with strong thermodynamic performance, making them attractive for premium residential hydronic systems and some larger commercial applications where manufacturers can solve flammability-related design and installation constraints.^{279 277}
- Case example: Stiebel Eltron's hpnex platform is a clear market-ready example of the transition, with the company stating that the new range works consistently with natural refrigerant R290 across air-source, ground-source, and integral ventilation applications, and that it was engineered around a new refrigeration circuit and safety concept.^{277 281}
- Competitive implication: The refrigerant shift is not only a compliance issue. It raises redesign costs, changes installer training needs, and can favor manufacturers with stronger in-house engineering, certification capability, and channel support, especially in Europe's residential and light-commercial monobloc and hydronic segments.^{276 284}
- Industrial relevance: For industrial heat pumps, refrigerant choice is increasingly linked to temperature lift, safety classification, and process integration rather than only residential eco-label positioning. JRC indicates that advanced industrial systems in the 150 to 200°C range are under development, which implies continued experimentation with working fluids and cycle architectures beyond the mainstream residential refrigerant debate.^{278 24}

7.2

System Integration and Renewable Energy

Innovation is increasingly occurring at system level rather than at the standalone appliance level. Heat pumps are being integrated more deeply with rooftop solar PV, solar thermal, thermal storage, shallow geothermal resources, district heating and cooling networks, and smart-grid control layers, which broadens their value proposition from efficient heating device to flexible energy-system asset.^{276 264 52}

Title — Integration pathways in heat pump innovation

INTEGRATION PATHWAY	MAIN TECHNICAL BENEFIT	STRATEGIC RELEVANCE
Solar PV plus heat pump	Improves self-consumption and supports lower operating cost.	Most scalable in residential and small commercial buildings.
Solar thermal plus heat pump	Raises source or pre-heat temperature and can reduce compressor work in some configurations.	More niche, but relevant in hybrid building and district systems.
Ground or geothermal plus heat pump	Stabilizes source temperature and improves seasonal performance.	Strong fit for premium residential, campuses, and networked systems.
District heating plus large heat pump	Enables waste-heat recovery, thermal storage, and utility-scale electrification.	High relevance for commercial, municipal, and industrial decarbonization.

Source: ^{264 52 50}

- Solar-coupled systems: A systematic review of solar-assisted ASHP systems found PV-assisted ASHP configurations delivered the strongest average techno-economic performance among the reviewed architectures, with mean COP around 3.75 versus about 2.90 for solar-thermal-assisted ASHP and 3.03 for PV/T-assisted ASHP systems.²⁶⁴
- District heating integration: Recent arXiv modelling on district systems shows that combining heat pumps with solar thermal and storage can materially change optimal system design, cost, and emissions outcomes, even if the exact optimum remains highly dependent on carbon pricing, source temperatures, and storage losses.⁵⁰
- Geothermal and networked thermal systems: Research on shallow geothermal potential in western Switzerland suggests that seasonal regeneration from cooling loads can materially increase GSHP technical potential, with individual GSHPs covering up to 55% of heating demand in the case study and district heating and cooling configurations raising coverage toward about 85%.⁵²
- Smart-grid functionality: The integration trend increasingly includes predictive controls, tariff response, and demand shifting. This matters because the economic value of a heat pump is rising when it can optimize around weather forecasts, occupancy, and electricity prices rather than simply maximize instantaneous COP.²⁶³
- Segment implication: Residential integration is led by PV plus battery plus heat pump packages, commercial integration by building management systems and thermal storage, and industrial or district applications by waste-heat recovery and central plant optimization. This reinforces that innovation is broad-based across residential, commercial, and industrial segments rather than confined to household equipment alone.^{276 278}

7.3

Efficiency Improvements and Consumer Benefits

Performance gains are increasingly being delivered through better controls, improved compressors and inverters, optimized refrigerant circuits, and system designs that preserve output in colder climates while reducing acoustic nuisance. For consumers and building owners, the most visible benefits are higher seasonal efficiency, lower backup-heating use, quieter operation, and improved suitability for retrofit markets in Northern and Eastern Europe.^{276 263 268}

- SCOP progression: Public market evidence from EPREL-based brand comparisons in earlier sections already showed average SCOP values clustering in the low-to-high 4s for leading European brands, and the technology trend is now toward preserving those seasonal gains under more demanding real-world conditions rather than only improving laboratory ratings.^{191 276}
- Cold-climate performance: Field evidence from a predictive-control demonstration in an all-electric house showed operation at outdoor temperatures down to minus 15°C, with average daily heating energy use reduced by 19% and backup electric resistance use reduced by 38%, indicating that software and controls are becoming as important as hardware in extending viable cold-climate performance.²⁶³
- Noise reduction: Acoustic performance is becoming a first-order adoption variable in dense European housing markets. A 2025 arXiv study on model predictive control for heat pump noise mitigation found that control-based strategies can reduce noise nuisance with negligible energy-cost increase, suggesting that future differentiation will come partly from operating intelligence rather than only from enclosure design.²⁶⁸
- Product-level signal: Stiebel Eltron states that its hpnex R290 range sets new standards in operation and noise levels as well as efficiency, illustrating how manufacturers are now marketing acoustic performance and user experience alongside refrigerant compliance and energy ratings.^{277 281}
- Industrial frontier: Efficiency innovation is also expanding upward in temperature. JRC reports that industrial heat pumps below 140°C have reached high technology-readiness levels, while advanced systems for 150 to 200°C are under development, which is strategically important because it expands the addressable share of industrial process heat that can be electrified.^{24 278}
- Investor takeaway: In residential markets, higher SCOP, lower noise, and stronger cold-weather output improve conversion and reduce subsidy dependence at the margin. In commercial and industrial markets, the bigger value driver is the combination of efficiency with controllability, higher supply temperatures, and integration into broader thermal and electrical systems.^{276 263 278}

8

Barriers to Adoption and Market Risks

8

Barriers to Adoption and Market Risks

Despite the 2025 rebound in annual shipments, Europe's heat pump market still faces structural barriers that can slow both near-term sales and longer-term installed-base expansion. The main risks are not technological but economic, labour-related, and policy-driven: distorted electricity-to-gas price signals, high upfront costs, installer shortages, stop-start subsidy design, and uneven consumer readiness continue to constrain adoption across residential, commercial, and industrial applications.^{16 11 113}

For investors, these barriers matter because they affect annual shipments first, but can also delay the growth of the installed base by weakening conversion rates, stretching project lead times, and reducing confidence in manufacturing and channel expansion. The risk profile is especially acute in gas-oriented markets such as Germany, Italy, and the UK, while Nordic markets show that stronger electricity economics and higher consumer familiarity can materially reduce adoption friction.^{314 113}

8.1

Economic Barriers and Tax Implications

Heat pump economics remain highly sensitive to the retail electricity-to-gas price ratio, because operating-cost competitiveness depends more on tariff design and taxation than on equipment efficiency alone. The IEA states that heat pumps can reduce bills where electricity and gas prices are comparable, but can lead to higher bills where electricity prices exceed gas prices by more than three times; its 2025 EU cost-competitiveness chart also shows that many member states still sit above the threshold at which electrification becomes broadly cost-competitive against fossil alternatives.^{303 314}

- Price-ratio barrier: EHPA argues that heat pumps are most competitive when electricity prices are no more than about twice gas prices, while the user-specified market benchmark of below roughly 2.5 remains directionally consistent with IEA cost-competitiveness analysis; in many European markets, however, the ratio remains above 3, materially weakening the running-cost case.^{113 303}
- Tax distortion: The IEA and EHPA both identify taxes, levies, and non-energy charges on electricity as a core barrier, because they often fall more heavily on power than on fossil gas even though electrification is the policy objective. This directly undermines residential payback, but it also weakens commercial and industrial business cases where electricity is the dominant operating input.^{16 314 113}
- Upfront capital cost: The Commission lists high installation cost, long customer journeys, and limited financing as major reasons for slower uptake in several national markets. In practice, installed heat pump systems still commonly cost around two to three times as much as gas boiler replacements in many retrofit cases, especially where emitter upgrades, electrical works, or envelope improvements are also required; this ratio is widely cited in market commentary, but public Europe-wide official datasets do not harmonise it precisely by country or segment.^{11 16}
- Segment effect: Residential adoption is most exposed to retail tariff distortions and household financing constraints, while commercial and industrial projects are more exposed to hurdle-rate sensitivity, power-price volatility, and uncertainty over future carbon and electricity pricing. That means poor electricity economics can suppress annual shipments across all segments even when the long-run installed-base case remains intact.^{16 290}

- Country contrast: Nordic markets demonstrate that stronger structural electricity economics reduce dependence on heavy capital subsidies, whereas gas-centric markets require larger grants just to neutralise distorted operating-cost signals. This is one reason penetration remains much higher in Norway and Finland than in the UK or much of Central and Eastern Europe, despite broad awareness of the technology.^{113 11}
- Investor implication: Unless governments rebalance taxes and levies away from electricity and toward fossil fuels or carbon, annual shipments will remain more cyclical and subsidy-dependent than the decarbonisation narrative implies. That raises demand volatility risk for manufacturers, distributors, and project developers, particularly in residential-heavy markets.^{314 113}

8.2

Workforce and Skill Gaps

Labour availability is now a binding market constraint rather than a secondary execution issue. Public estimates vary, but the most credible current range is that Europe needs roughly 500,000 to 750,000 additional skilled workers by 2030, with EHPA specifically citing a minimum need for 500,000 skilled full-time-equivalent employees and JRC noting that around 50% of existing installers will need reskilling.^{306 38 311}

Title — Europe heat pump workforce gap indicators

INDICATOR	LATEST PUBLIC FIGURE	MARKET IMPLICATION
Additional skilled workers needed by 2030	500,000 minimum	Large labour gap versus current installation capacity.
Industry estimate range for additional workers	500,000 to 750,000	Planning assumptions vary, but shortage is material in all cases.
Existing installers needing reskilling	50%	Transition requires retraining, not only new entrants.
Training courses mapped by EU platform	50+	Training ecosystem exists, but scale-up remains incomplete.

Source: ^{306 38 312}

- Installation bottleneck: The Commission explicitly identifies installer shortages and long waiting periods as barriers to deployment. This affects annual shipments directly because demand cannot convert into completed installations at the pace implied by policy targets, even when subsidies are available.¹¹
- Skill complexity: The labour challenge is widening because the market is moving toward hydronic retrofits, natural refrigerants such as R290, digital controls, and larger commercial and industrial systems. These require broader competencies than legacy boiler replacement, including electrical integration, refrigerant safety, commissioning, controls optimisation, and in some cases process-heat engineering.^{311 312}
- Reskilling need: JRC’s estimate that about half of existing installers require reskilling is strategically important because the bottleneck is not solved by recruitment alone. Europe must convert parts of the incumbent heating workforce from fossil-based systems to electrified thermal systems while also expanding the total labour pool.³⁸

- Segment coverage: Workforce shortages are often discussed in residential terms, but the constraint is broader. Commercial and industrial heat pumps require design engineers, controls specialists, refrigeration technicians, and project integrators, so labour scarcity can disproportionately delay higher-value projects even if unit volumes are lower than in residential markets.^{311 11}
- Training response: EHPA reports that the Heat Pump Accelerator platform is mapping more than 50 training courses across EU member states, while projects such as RESKILL4NETZERO, SKILLSAFE EU, and HEATCRAFTHP are building scalable training and safety frameworks. These are constructive signals, but they remain early relative to the labour intensity required to move annual installations toward 4 million by 2030.^{305 312 311}
- Investor implication: Labour scarcity raises execution risk across the value chain by extending lead times, increasing installation costs, and constraining service quality. It also favours manufacturers and distributors with captive installer networks, training academies, or strong channel partnerships, because workforce access becomes a competitive moat rather than a back-office function.^{11 312}

8.3

Policy and Consumer Awareness Challenges

Policy volatility remains one of the clearest explanations for the 2023 to 2024 market downturn and for the uneven 2025 recovery. The Commission attributes weak adoption in several markets to unstable financing, complex customer journeys, and policies that in some cases discourage deployment, while EHPA links the prior market collapse directly to abrupt policy changes and calls for long-term regulatory certainty.^{11 113}

- Subsidy unpredictability: Italy's Superbonus wind-down and repeated redesigns of German heating support are the clearest recent examples of stop-start policy risk. Even when support remains generous in headline terms, sudden changes in eligibility, timing, or administration can freeze consumer decisions and disrupt installer order books.^{113 11}
- Demand volatility mechanism: Policy instability affects annual shipments immediately because households and project developers delay commitments when grant rules are unclear. Installed-base growth is then affected with a lag, as weaker annual additions reduce future replacement, service, and ancillary revenue pools.^{11 16}
- Consumer awareness gap: The Commission highlights the long and complex customer journey as a barrier, and recent behavioural research in five EU countries found that performance expectancy, price value, and effort expectancy are among the strongest predictors of adoption intent for heating-related smart technologies. While that study is not heat-pump-specific, it supports the inference that perceived complexity and weak understanding of value remain important barriers in heating technology adoption more broadly.^{11 291}
- Cultural inertia: Gas-heating markets face a deeper behavioural barrier than high-penetration Nordic markets because incumbent technologies are familiar, installer networks are entrenched, and consumers often perceive heat pumps as complex or risky in retrofit settings. This cultural factor is difficult to quantify directly, but it is consistent with the Commission's emphasis on information campaigns and with cross-country differences in adoption despite similar decarbonisation goals.^{11 113}
- Information quality risk: In lower-penetration markets, poor installer advice, misinformation, and confusion over building suitability can damage trust and reduce conversion. This is especially relevant in Central and Eastern Europe and in gas-oriented Western markets where first-time adopters have less peer evidence and less familiarity with system design requirements.^{11 311}

- Strategic conclusion: The market does not only need more subsidies; it needs more predictable subsidies, simpler customer journeys, and stronger public understanding of where heat pumps work economically and technically. Without those conditions, Europe can still grow its installed base, but annual shipments are likely to remain volatile and below the pace required for 2030 targets across residential, commercial, and industrial segments.^{113 16 11}

Market Forecasts and Strategic Outlook to 2030

9

Market Forecasts and Strategic Outlook to 2030

The 2030 outlook remains structurally positive, but the market is still below the deployment pace implied by EU ambitions. Market value is projected to rise from USD 16.1 billion in 2025 to about USD 31.1 billion by 2030 at a 14.1% CAGR, while annual installations need to move from roughly 2.34 million to 2.9 million in 2025, depending on dataset scope, toward around 4 million per year by 2030 under the European Commission’s current KPI framework and potentially closer to 6 million per year under broader REPowerEU-style deployment logic.^{58 112 325 330}

The key strategic issue is the gap between cumulative stock ambition and annual flow reality. Europe’s installed base was around 28 million to 29.3 million units in 2025, so reaching 60 million by 2030 implies roughly 30 million additional units over five years, which is materially above the recent annual shipments run rate even before accounting for retirements and uneven segment mix across residential, commercial, and industrial applications.^{112 331 73}

9.1

Market Value Trajectory and Target Installations

The value case to 2030 is robust even if unit deployment remains volatile. MarketsandMarkets projects the European heat pump market to expand from USD 16,113.6 million in 2025 to USD 31,115.5 million by 2030, implying a 14.1% CAGR, which is consistent with a market that can still nearly double in value despite policy-driven swings in annual shipments.⁵⁸

Title — Europe heat pump outlook to 2030

METRIC	2025 BASELINE	2030 TARGET OR FORECAST
Market value, USD bn	16.1	31.1
Installed base, million units	28.0 to 29.3	60.0 ambition
Annual installations, million units	2.34 to 2.90	4.0 KPI
Implied annual additions for 60m stock goal, million units per year	Below target today	About 5.0 to 6.0

Source: ^{58 112 325 330 331}

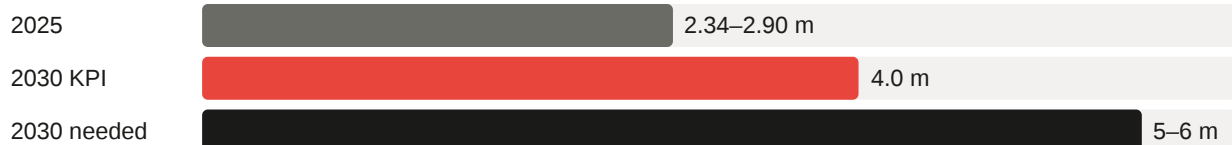
MARKET VALUE (USD BN)



INSTALLED BASE (MILLION UNITS)



ANNUAL INSTALLATIONS (MILLION UNITS / YEAR)



■ 2025 baseline ■ 2030 target / forecast ■ 2030 implied need (60 m stock goal)

- Installed base versus annual shipments: The 60 million figure is a cumulative stock ambition, not an annual sales target. By contrast, the Commission's 4 million figure is an annual installation KPI for 2030, so the two metrics should not be treated as interchangeable.^{112 325}
- Feasibility gap: Starting from roughly 28 million to 29.3 million installed units in 2025, a 60 million stock by 2030 requires around 30 million additional units over five years, or about 6 million gross additions per year before retirements. That is well above the 2025 annual flow of 2.34 million to 2.9 million units and still above the Commission's 4 million 2030 KPI, implying that current policy targets are directionally supportive but not yet fully sufficient for the higher stock ambition.^{112 73 331 330}
- Momentum signal: Q1 2026 residential sales rose 17% year on year across 11 European countries, which is a constructive leading indicator after the 2024 downturn and 2025 rebound. However, it remains a partial residential sample rather than a full-year all-segment Europe total, so it supports recovery but does not yet prove a trajectory consistent with 60 million installed units by 2030.^{112 328}
- System integration requirement: arXiv-based power-system studies indicate that a large-scale heat pump rollout by 2030 is feasible only if accompanied by additional wind and solar capacity plus thermal storage. One study finds that meeting 25% of building heat demand with heat pumps in 2030 is best supported by additional wind capacity, while another finds long-duration thermal storage attached to district heating can reduce long-duration electricity storage needs by an average of 36%.^{4 49}
- Segment implication: Residential will continue to dominate unit additions, but commercial, district, and industrial heat pumps will matter disproportionately for value growth and gas displacement. That means the market can hit value forecasts without fully meeting stock ambitions if mix shifts toward larger and more complex systems, but such an outcome would still leave Europe short of its broader electrification and energy-security goals.^{58 16 11}

9.2

Strategic Impacts and Geopolitical Considerations

Heat pumps are no longer only a decarbonisation technology. They are increasingly being treated as an energy-security asset because they reduce exposure to imported gas, improve resilience

against fuel-price shocks, and support a broader electrification strategy tied to domestic renewables and European industrial policy.^{119 16 325}

- Gas security: The user-specified framing that Russian gas dependence fell from about 45% to around 12% is directionally consistent with current official energy-security discourse, but the most reliable current source located for this section is ACER's statement that Russian gas still accounts for around 12% of EU gas demand in 2026. IEA also reports that Russian piped gas deliveries to the EU fell 45% year on year in the first half of 2025 after the halt of transit via Ukraine, reinforcing the strategic logic for faster heating electrification.^{327 326}
- Geopolitical hedge: The Commission's 2026 Electrification Action Plan explicitly links heat pumps to competitiveness, energy security, and lower fossil fuel dependence, including measures to improve the electricity-to-gas price ratio and mobilise financing through the Heat Pump Accelerator Platform. This makes future market growth more sensitive to macro-energy policy than to appliance policy alone.^{325 124}
- Price-shock sensitivity: The Q1 2026 rebound was partly associated with higher gas and oil prices after renewed Middle East disruption, according to market reporting cited by pv magazine. That suggests annual shipments can accelerate quickly when fossil-fuel risk becomes salient, but it also underlines that Europe still lacks a stable structural price framework and remains too dependent on crisis-driven demand spikes.^{328 11}
- Power-system consequence: Large-scale electrification shifts geopolitical exposure from imported gas toward domestic grid adequacy, renewable build-out, and critical-component supply chains. arXiv modelling shows that simultaneous European heat pump rollout raises winter peak electricity needs and is best paired with more wind generation and thermal storage, while broader European gas-independence modelling finds that cutting gas reliance is economically compelling but does not fully shield consumers from price volatility unless electricity-market reforms also follow.^{4 49 290}
- Industrial and district angle: Geopolitical resilience is not only a residential story. IEA highlights large-scale heat pumps in district heating and industry as strategic components of long-term system transformation, which matters because these applications can displace gas in municipal networks and process heat with fewer unit installations but larger energy impact per project.¹⁶
- Strategic conclusion: By 2030, the strongest markets are likely to be those that combine heat pump deployment with electricity-price reform, renewable expansion, thermal storage, and local manufacturing support. Markets that rely only on consumer subsidies will remain more cyclical and more exposed to geopolitical fuel-price shocks.^{325 112 290}

9.3

Recommendations for Key Stakeholders

The strategic priority through 2030 is to convert a cyclical recovery into a durable scale-up. That requires different actions for manufacturers, investors, and policymakers, but all three groups need to plan around the distinction between installed base growth as the long-run stock objective and annual shipments as the near-term execution metric.^{112 73}

- Manufacturers: Prioritise capacity additions in hydronic residential, light commercial, district, and industrial platforms rather than only mass-market air-source volume. The most resilient strategies will combine European assembly with deeper localisation of compressors, inverters, and controls, while using M&A and installer-network partnerships to secure route-to-market and service capacity.^{16 333 49}
- Manufacturers: Increase R&D spending on low-GWP refrigerants, cold-climate performance, noise reduction, and flexible operation with thermal storage. These features improve competitiveness

not only in residential retrofits but also in commercial and district applications where grid interaction and operating-cost optimisation will increasingly determine project bankability.^{11 16 4}

- Investors: Allocate capital across three buckets rather than treating the market as a single residential hardware story. The most attractive mix is likely to combine scale exposure in France, Germany, Italy, and the UK; premium and technology-validation exposure in Nordic markets; and selective growth optionality in lower-penetration Central and Eastern Europe where policy execution improves.^{11 112 73}
- Investors: Favour business models with recurring revenue and policy resilience, including service networks, controls, thermal storage, district-heating integration, and industrial process-heat developers. These areas are less exposed to pure consumer subsidy cycles and better aligned with the Commission's electrification and energy-security agenda.^{325 16}
- Policymakers: Treat the 4 million annual installation KPI as a minimum waypoint, not the end-state, if the 60 million installed-base ambition is to remain credible. That implies multi-year subsidy visibility, faster permitting, electricity-tax and network-charge reform, and explicit support for thermal storage, district heating, and industrial heat pumps rather than a residential-only policy lens.^{112 325 330}
- Policymakers: Expand workforce and data infrastructure in parallel with demand support. Public sources still do not provide harmonised Europe-wide segment shipment data, and the labour bottleneck remains material, so better reporting and faster installer training are necessary to convert policy ambition into actual annual flow growth across residential, commercial, and industrial segments.^{11 112 16}
- Bottom line: The base case to 2030 is continued value expansion with uneven unit growth. The upside case requires Europe to move from subsidy-led recovery to system-led scale-up, while the downside case is a repeat of 2023 to 2024 style policy volatility that keeps annual shipments below the level needed to close the gap between today's installed base and the 60 million stock ambition.^{58 11 73}

10

References and Citations

10

References and Citations

This section consolidates the source base used across the study and explains how citations were selected, prioritised, and reconciled. The reference architecture was built around primary institutional datasets for installed base, annual shipments, policy, manufacturing, and technology, then supplemented with specialised market intelligence, technical studies, and company or regulator disclosures where primary public data were incomplete or more granular evidence was required.^{10 11 16 24}

10.1

Primary Data Sources

- European Heat Pump Association: Core source for Europe-wide annual shipments, installed base, country rankings, and market trend commentary. The study relied especially on EHPA Market Data, the March 2026 "Pump it up" note, the 2024 market-statistics executive summary, and selected press releases for 2024 to 2026 trend updates.^{10 12 23 17}
- International Energy Agency: Used for cross-country synthesis, energy-security implications, electrification economics, manufacturing outlook, and technology deployment context. Key references included Heat Pump Monitor 2026, Global Energy Review 2026 technology notes, Electricity 2026, Gas Market Report Q3 2025, and Energy Technology Perspectives 2026.^{16 13 314 326 255}
- European Commission and affiliated EU platforms: Primary source set for EU policy, accelerator governance, 2030 installation targets, electrification strategy, and official market framing. This included the Commission heat-pumps portal, Heat Pump Accelerator Platform pages, AccelerateEU materials, Electrification Action Plan documentation, Innovation Fund heat-auction material, and EUR-Lex legal texts.^{11 40 119 124 325 120}
- Joint Research Centre and SETIS: Primary technical and industrial evidence base for EU manufacturing footprint, value chains, domestic content, industrial heat pumps, technology readiness, and country fiches. The most important references were the 2025 EU heat pump status report, earlier JRC market and value-chain studies, industrial heat-pump analyses, and SETIS-hosted status reporting.^{24 260 38 60}
- EPREL and EU product databases: Used as the underlying public registry context for model-count and efficiency comparisons, with EPREL itself treated as the primary registry and third-party analytics treated as derived interpretations.²⁰⁰
- National and regulatory primary sources: Used where country policy design, subsidy levels, or tax treatment required official confirmation. These included KfW, France Rénov', ADEME, Ofgem, UK legislation, Enova, Vero, GOV.UK, and ACER.^{137 160 164 133 145 162 159 100 327}
- ArXiv and open technical papers: Used selectively for system-modelling, storage, geothermal integration, controls, noise mitigation, and behavioural or power-system implications where institutional market datasets did not provide technical depth. These were supporting technical sources rather than primary market-count sources.^{4 49 1 179 263 268}

10.2

Supplementary References

- Commercial market-sizing and forecast providers: MarketsandMarkets was used for total market value, segment value, and CAGR framing because public institutional sources did not provide a directly comparable Europe-wide revenue forecast split across residential, commercial, and industrial segments. Grand View Research was used only as a secondary directional check on application mix, not as the lead source for core market totals.^{58 65}
- BUILD UP and Commission dissemination pages: Used as secondary interpretive sources that often republished or contextualised EHPA and Commission findings in a policy-relevant format. These were particularly useful for country-by-country narrative synthesis and cost-reduction commentary.^{14 101 88}
- CE Delft and related industrial-policy studies: Used to benchmark domestic-content estimates, supply-chain bottlenecks, and component vulnerabilities, especially for compressors, electronics, and low-technology chips.²⁰⁴
- HouseHeating Pulse and Grid and Degrees: Used as derived analytics layers for subsidy comparisons, EPREL-based brand-share rankings, SCOP snapshots, and cross-country policy dashboards. These sources were treated as secondary and were only used where their underlying basis was transparent and consistent with official registries or public policy documents.^{83 191 190 85}
- Trade press and specialist media: pv magazine was used sparingly for timely reporting on Q1 2026 sales momentum and market reaction to energy-price movements, with preference given to stories directly attributing figures to EHPA or Commission sources.³²⁸
- Company and transaction disclosures: Used for product launches, manufacturing localisation, and M&A verification where official company or competition filings were available. These included Stiebel Eltron, Midea, Hisense, NIBE, Danfoss, and European Commission merger documents.^{277 281 209 213 212 211 206 221}
- Additional institutional and advisory references: Used where they added country-specific or sector-specific context, especially for UK progress tracking and Italian policy interpretation.^{102 177}
- Scope note: Some supplementary sources are interpretive, commercial, or advocacy-oriented rather than statistical authorities. They were therefore used to extend coverage, not to override EHPA, IEA, European Commission, JRC, EPREL, or national regulator data where those primary sources existed.^{10 16 11 24}

10.3

Citation Methodology

- Source hierarchy: Citations were prioritised in descending order of evidentiary weight: official EU institutions and regulators, EHPA and IEA, JRC and SETIS technical reports, national government or regulator documents, EPREL registry material, peer-reviewed or open technical papers, then transparent secondary analytics and specialist media. This hierarchy was used consistently to minimise reliance on commercial or interpretive sources where primary evidence was available.^{11 16 24}
- Stock versus flow discipline: Installed base and annual shipments were cited separately and never merged into a single market-size figure. Where sources used different geographies or product scopes, the citation text explicitly stated whether the figure referred to 21-country Europe, 16-country EHPA coverage, 13-country EU preliminary data, EU-only stock, residential subsets, or Q1 2026 partial samples.^{10 12 11}

- Reconciliation rule: When multiple credible sources reported different figures, the study preserved the range and explained the reason for divergence rather than forcing a single synthetic number. EHPA was generally used for broad Europe shipment and installed-base headlines, the Commission for EU policy framing and preliminary member-state tracking, and IEA for synthesis and interpretation.^{10 11 13}
- Segment methodology: Because public Europe-wide unit shipments by residential, commercial, and industrial segment are not harmonised, segment analysis relied on market value forecasts, application evidence, industrial process-heat studies, and project-level or technology-readiness sources. This limitation was flagged explicitly wherever segment unit precision could not be supported.^{58 60 38}
- Manufacturer methodology: EPREL model counts were cited as model-registration or product-breadth indicators, not as direct shipment share. Where third-party EPREL analytics were used for brand ranking or SCOP averages, the report stated that these were proxies and not audited annual sales shares.^{200 191}
- Timeliness rule: For policy, subsidies, Q1 2026 sales, and 2026 regulatory developments, only sources current to 2025 to 2026 were used where possible. Older sources were retained only for historical baselines, structural context, or technical studies that remained relevant and were not superseded by newer institutional evidence.^{40 119 328}
- Transparency rule: Where evidence was partial, inferred, or commercially sourced, the report labelled it accordingly. Examples include market-value forecasts from commercial providers, policy interpretation from secondary dashboards, and technical extrapolations from arXiv system studies.^{58 83 4}
- Formatting rule: Citations were attached directly to factual claims, tables used a dedicated source line, and URLs were kept at page level rather than domain level to preserve traceability. This approach was designed to allow an industry reader to verify each major statistic, policy statement, or competitive claim back to its originating document.^{10 24 11}



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